

Лабораторная работа №8

Планировщики событий

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Информация

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Цели и задачи

- Получение навыков работы с планировщиками событий cron и at.
1. Выполните задания по планированию задач с помощью crond (см. раздел 8.4.1).
 2. Выполните задания по планированию задач с помощью atd (см. раздел 8.4.2).

Материалы и методы

- Linux (дистрибутив Rocky 9.6)
- Linux Fedora Sway (Markdown)
- Oracle VirtualBox

Выполнение лабораторной работы

8.4.1. Планирование задач с помощью cron

su -

systemctl status crond -l

cat /etc/crontab

```
[aeakunaeva@aeakunaeva ~]$ su -
Password:
[root@aeakunaeva ~]# systemctl status crond -l
● crond.service - Command Scheduler
   Loaded: loaded (/usr/lib/systemd/system/crond.service; enabled; preset: en>
   Active: active (running) since Sat 2025-12-20 17:05:10 MSK; 2h 12min ago
     Main PID: 998 (crond)
        Tasks: 1 (limit: 22987)
       Memory: 1.2M
          CPU: 82ms
      CGroup: /system.slice/crond.service
              └─998 /usr/sbin/crond -n

Dec 20 17:05:10 aeakunaeva.localdomain systemd[1]: Started Command Scheduler.
Dec 20 17:05:10 aeakunaeva.localdomain crond[998]: (CRON) STARTUP (1.5.7)
Dec 20 17:05:10 aeakunaeva.localdomain crond[998]: (CRON) INFO (Syslog will be >
Dec 20 17:05:10 aeakunaeva.localdomain crond[998]: (CRON) INFO (RANDOM_DELAY wi>
Dec 20 17:05:10 aeakunaeva.localdomain crond[998]: (CRON) INFO (running with in>
Dec 20 18:01:01 aeakunaeva.localdomain CROND[2956]: (root) CMD (run-parts /etc/>
Dec 20 18:01:01 aeakunaeva.localdomain CROND[2955]: (root) CMDEND (run-parts /e>
Dec 20 19:01:01 aeakunaeva.localdomain CROND[3003]: (root) CMD (run-parts /etc/>

[root@aeakunaeva ~]# cat /etc/crontab
SHELL=/bin/bash
PATH=/sbin:/bin:/usr/sbin:/usr/bin
MAILTO=root

# For details see man 4 crontab
```

Отображение и изменение расписания в cron

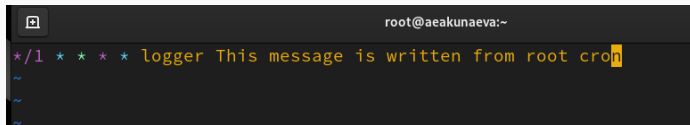
```
crontab -l
```

```
crontab -e
```

```
*/1 * * * * logger This message is written from root cron
```

```
# | | | .----- month (1 - 12) OR jan,feb,mar,apr ...
# | | | | .---- day of week (0 - 6) (Sunday=0 or 7) OR sun,mon,tue,wed,thu,fri,sat
# | | | | |
# * * * * * user-name  command to be executed

[root@aeakunaeva ~]# crontab -l
no crontab for root
[root@aeakunaeva ~]# crontab -e
no crontab for root - using an empty one
crontab: installing new crontab
```



A terminal window with a dark background. The title bar shows a window icon and the text "root@aeakunaeva:~". The terminal content shows a cron job entry: "* /1 * * * * logger This message is written from root cron". The asterisks are colored: the first is purple, the next three are green, and the last is purple. The text "logger" is orange, and the rest of the line is yellow. The cursor is at the end of the line. Below the entry are three tilde symbols (~) on separate lines.

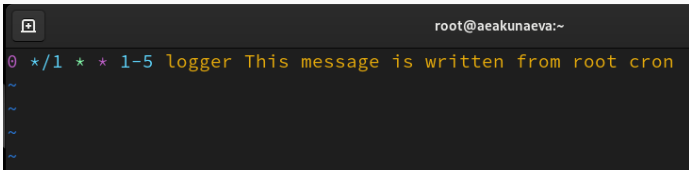
```
root@aeakunaeva:~  
*/1 * * * * logger This message is written from root cron  
~  
~  
~
```

Внесение задач в расписание cron

```
[root@aeakunaeva ~]# crontab -l
*/1 * * * * logger This message is written from root cron
[root@aeakunaeva ~]# grep written /var/log/messages
Dec 20 19:23:02 aeakunaeva root[3134]: This message is written from root cron
Dec 20 19:24:01 aeakunaeva root[3160]: This message is written from root cron
Dec 20 19:25:01 aeakunaeva root[3183]: This message is written from root cron
[root@aeakunaeva ~]# crontab -e
crontab: installing new crontab
[root@aeakunaeva ~]# crontab -l
0 */1 * * 1-5 logger This message is written from root cron
[root@aeakunaeva ~]# cd /etc/cron.hourly
[root@aeakunaeva cron.hourly]# touch eachhour
[root@aeakunaeva cron.hourly]# nano eachhour
[root@aeakunaeva cron.hourly]# chmod +x eachhour
[root@aeakunaeva cron.hourly]# cd /etc/cron.d
[root@aeakunaeva cron.d]# touch eachhour
[root@aeakunaeva cron.d]# nano eacchour
[root@aeakunaeva cron.d]# nano eachhour
[root@aeakunaeva cron.d]# grep written /var/log/messages
Dec 20 19:23:02 aeakunaeva root[3134]: This message is written from root cron
Dec 20 19:24:01 aeakunaeva root[3160]: This message is written from root cron
Dec 20 19:25:01 aeakunaeva root[3183]: This message is written from root cron
Dec 20 19:26:02 aeakunaeva root[3210]: This message is written from root cron
[root@aeakunaeva cron.d]#
```

Внесение новой записи в cron

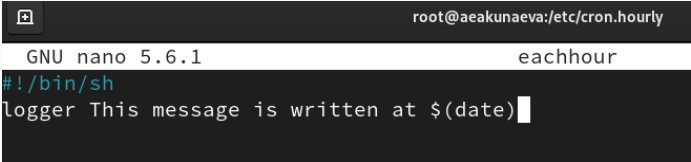
```
0 */1 * * 1-5 logger This message is written from root cron
```

A terminal window with a dark background. The title bar shows a window icon and the text 'root@aeakunaeva:~'. The terminal content shows the cron job entry '0 */1 * * 1-5 logger This message is written from root cron' in a yellow monospace font. Below the entry, there are four blue tilde characters '~' on separate lines.

```
root@aeakunaeva:~  
0 */1 * * 1-5 logger This message is written from root cron  
~  
~  
~  
~
```

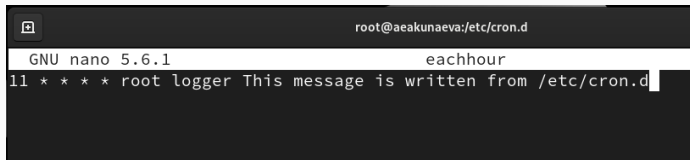
Изменение файла сценария /etc/cron.hourly/eachhour

```
#!/bin/sh  
logger This message is written at $(date)
```

A screenshot of a terminal window showing the nano text editor. The title bar at the top indicates the user is root at aeakunaeva, editing the file /etc/cron.hourly/eachhour. The editor status line shows 'GNU nano 5.6.1' on the left and 'eachhour' on the right. The main editing area contains the same two lines of code as the previous block: a shebang line and a logger command. A white cursor is positioned at the end of the second line.

```
root@aeakunaeva:/etc/cron.hourly  
GNU nano 5.6.1 eachhour  
#!/bin/sh  
logger This message is written at $(date)
```

```
11 * * * * root logger This message is written from /etc/cron.d
```



```
root@aeakunaeva:/etc/cron.d
GNU nano 5.6.1 eachhour
11 * * * * root logger This message is written from /etc/cron.d
```

8.4.2. Планирование заданий с помощью at

```
[aeakunaeva@aeakunaeva ~]$ su -
Password:
[root@aeakunaeva ~]# systemctl status atd
• atd.service - Deferred execution scheduler
   Loaded: loaded (/usr/lib/systemd/system/atd.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-12-20 17:05:10 MSK; 2h 30min ago
     Docs: man:atd(8)
   Main PID: 997 (atd)
      Tasks: 1 (limit: 22987)
     Memory: 896.0K
        CPU: 8ms
    CGroup: /system.slice/atd.service
            └─997 /usr/sbin/atd -f

Dec 20 17:05:10 aeakunaeva.localdomain systemd[1]: Started Deferred execution sched>

[root@aeakunaeva ~]# at 19:37
warning: commands will be executed using /bin/sh
at> logger message from at
at> <EOT>
job 3 at Sat Dec 20 19:37:00 2025
[root@aeakunaeva ~]# atq
1          Sun Dec 21 19:30:00 2025 a root
2          Sun Dec 21 19:32:00 2025 a root
3          Sat Dec 20 19:37:00 2025 a root
[root@aeakunaeva ~]# grep 'from at' /var/log/messages
Dec 20 20:36:40 aeakunaeva root[3554]: message from at
[root@aeakunaeva ~]#
```


Выводы

Я получила навыки работы с планировщиками событий cron и at.